Technical University of Crete

School of Electrical and Computer Engineering

Course: Reinforcement Learning and Dynamic Optimization

Programming Assignment 1

Environment setup

The simulation environment is initialized using the random seed 63 in order to ensure reproducibility. For each user $i \in \{1, 2\}$ and channel $j \in \{1, \ldots, K\}$, a baseline signal-to-noise ratio $\mathrm{SNR}_{i,j}$ is generated uniformly in the interval [0.8, 2.0]:

$$\mathbf{SNR}_{i,j} = \begin{bmatrix} 1.47209525 & 1.99071064 & 1.03602485 & 1.40634513 & 1.04045493 \\ 1.00679373 & 1.24114376 & 1.1453754 & 1.29648329 & 1.98738394 \end{bmatrix}$$

At each round t, the instantaneous SNR is modeled as a random perturbation of the baseline value:

$$\mathrm{SNR}^t_{i,j} \sim \mathrm{Uniform}\Big(\mathrm{SNR}_{i,j} - \epsilon,\ \mathrm{SNR}_{i,j} + \epsilon\Big)\ ,\ \epsilon = 0.6$$

The instantaneous transmission rate of user i on channel j is then given by: $R^t_i(j) = \log\Big(1 + 10\,\mathrm{SNR}^t_{i,j}\Big)$.

Task 0: Modeling

Case A

An arm a is a pair of distinct channels (j_1, j_2), assigning channel j_1 to UE1 and j_2 to UE2, with $j_1 \neq j_2$. Therefore, the arm set is defined as $\mathcal{A} = \{(j_1, j_2) : j_1 \neq j_2\}$. Since in this case only the aggregate reward is observed, each pair $a_t = (j_{1,t},\ j_{2,t})$ is treated as an distinct bandit arm, with the learner observing the sum-rate: $r_t(a_t) = R^t_1(j_{1,t}) + R^t_2(j_{2,t})$.

Case B

The action remains the selection of a pair of distinct channels $(j_1,\ j_2)$, with $j_1 \neq j_2$. However, the learner now observes the individual rewards $R^t_1(j_1), R^t_2(j_2)$, rather than only their sum. Thus, the learner can estimate the expected rate per user-channel combination $\mu_{i,j} = \mathbb{E}\Big[R_i(j)\Big]$. Since the expected reward decomposes additively, $\mu_{(j_1,j_2)} = \mu_{1,j_1} + \mu_{2,j_2}$, the learner estimates rewards per user–channel component rather than per arm, reducing the number of unknown quantities from $K(K-1)$ to $2K$.

In addition, a common step across the two cases is the computation of the expected reward of each arm, which is obtained by taking the expectation of the reward function under the uniform distribution of the instantaneous SNR. The expected rate of UE i assigned with channel j at time t is,

$$\mathbb{E}[R^t_i(j)] = \mathbb{E}[\log(1 + 10\,\mathrm{SNR}^t_{i,j}] = \frac{1}{2\epsilon}\int_{\mathrm{SNR}_{i,j}-\epsilon}^{\mathrm{SNR}_{i,j}+\epsilon} \log(1 + 10s)\,ds$$

Using integration by parts,

$$\mathbb{E}[\log(1 + 10\,\mathrm{SNR}^t_{i,j}] = \frac{F(\mathrm{SNR}_{i,j} + \epsilon) - F(\mathrm{SNR}_{i,j} - \epsilon)}{2\epsilon},\ \text{where}\ F(s) = \frac{(1 + 10s)\,(\log(1 + 10s) - 1)}{10}$$

Now consider an arm $a_t = (j_{1,t}, j_{2,t})$. Its reward is: $r_t(a_t) = \log(1 + 10\,\mathrm{SNR}^t_{1,j_1}) + \log(1 + 10\,\mathrm{SNR}^t_{2,j_2})$. The expected reward is,

$$\mathbb{E}[r_t(a_t)] = \mathbb{E}[\log(1 + 10\,\mathrm{SNR}^t_{1,j_1}) + \log(1 + 10\,\mathrm{SNR}^t_{2,j_2})] = \mathbb{E}[\log(1 + 10\,\mathrm{SNR}^t_{1,j_1})] + \mathbb{E}[\log(1 + 10\,\mathrm{SNR}^t_{2,j_2})]$$

Using the result above,

$$\mathbb{E}[R^t_{j_1,j_2}] = \mu_{j_1,j_2} = \frac{F(\mathrm{SNR}_{1,j_1} + \epsilon) - F(\mathrm{SNR}_{1,j_1} - \epsilon)}{2\epsilon} + \frac{F(\mathrm{SNR}_{2,j_2} + \epsilon) - F(\mathrm{SNR}_{2,j_2} - \epsilon)}{2\epsilon}$$

Task 1: Implement UCB for Case A

Reward range. The reward obtained from any arm lies in the interval $r_t \in [2\log 3,\ 2\log 27]$, thus the reward range used in Hoeffding's inequality is: $R = \max r_t - \min r_t = 2\log 27 - 2\log 3 = 2\log 9$.

Choice of the UCB bonus term. The exploration bonus is chosen from Hoeffding's inequality. If the reward of an arm lies in an interval of length R, then for any arm a and any number of pulls $N_a(t)$,

$$\Pr(|\hat{\mu}_a(t) - \mu_a| > \epsilon) \le \exp\left(-\frac{2N_a(t)\epsilon^2}{R^2}\right)$$

We set the upper bound of the probability to T^{-4} and thus obtain: $\epsilon = R\sqrt{\frac{2\log T}{N_a(t)}}$

UCB Policy. For $t = 1, \ldots, T$:

- Each arm is selected once for $t \le |\mathcal{A}|$.
- For $t > |\mathcal{A}|$, select: $a_t = \arg\max_{a\in\mathcal{A}} \mathrm{UCB}_a(t)$, with $\mathrm{UCB}_a(t) = \hat{\mu}_a(t) + R\sqrt{\frac{2\log T}{N_a(t)}}$.
- Observe $r_t(a_t) = R^t_1(j_{1,t}) + R^t_2(j_{2,t})$ and update the statistics $N_{a_t}(t)$, $\hat{\mu}_{a_t}(t)$ of the selected arm a_t.
- Track regret: $\mathbb{E}[\mathrm{Reg}(t)] = \mathbb{E}\left[\sum_{s=1}^t (\mu^* - \mu_{a_s})\right], \mu^* = \max_{j_1\neq j_2} \mu_{j_1,j_2}$.
  By defining the suboptimality gap $\Delta_a = \mu^* - \mu_a$,

$$\mathbb{E}[\mathrm{Reg}(t)] = \mathbb{E}\left[\sum_{s=1}^t \Delta_{a_s}\right] = \mathbb{E}\left[\sum_a \Delta_a N_a(t)\right] = \sum_a \Delta_a \mathbb{E}[N_a(t)] = \sum_{a\neq a^*} \Delta_a \mathbb{E}[N_a(t)]$$

Calculating instance-dependent bound

Step 1: Good event. Define the good event $\mathcal{G} := \left\{\forall a, \forall t : |\hat{\mu}_a(t) - \mu_a| \le R\sqrt{\frac{2\log T}{N_a(t)}}\right\}$

Step 2: Pull bound on $\mathcal{G}$. Suppose arm $a \neq a^*$ is selected at round t. Then by UCB,

$$\hat{\mu}_a(t) + R\sqrt{\frac{2\log T}{N_a(t)}} \ge \hat{\mu}_{a^*}(t) + R\sqrt{\frac{2\log T}{N_{a^*}(t)}}$$

On $\mathcal{G}$, the estimate of a suboptimal arm cannot be too large, and at the same time the estimate of the optimal arm cannot be too small,

$$\hat{\mu}_a(t) \le \mu_a + R\sqrt{\frac{2\log T}{N_a(t)}}, \quad \hat{\mu}_{a^*}(t) \ge \mu^* - R\sqrt{\frac{2\log T}{N_{a^*}(t)}}$$

Therefore,

$$\mu_a + 2R\sqrt{\frac{2\log T}{N_a(t)}} \ge \mu^* \Leftrightarrow \Delta_a \le 2R\sqrt{\frac{2\log T}{N_a(t)}} \Leftrightarrow N_a(t) \le \frac{8R^2\log T}{\Delta_a^2}$$

Step 3: Probability of $\mathcal{B}$. $\Pr(\mathcal{B}) = \Pr\left(\exists\, a, t : |\hat{\mu}_a(t) - \mu_a| > R\sqrt{\frac{2\log T}{N_a(t)}}\right) \leq |\mathcal{A}|T^{-3}$

Step 4: Probability of $\mathcal{G}$. $\Pr(\mathcal{G}) = 1 - \Pr(\mathcal{B}) \geq 1 - |\mathcal{A}|T^{-3} \approx 1$

Step 5: Expected regret decomposition. $\mathbb{E}[\mathrm{Reg}(T)] = \Pr(\mathcal{G})\,\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{G}] + \Pr(\mathcal{B})\,\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{B}]$
On $\mathcal{G}$,

$$\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{G}] = \sum_{a \neq a^*} \Delta_a \mathbb{E}[N_a(T) \mid \mathcal{G}] \leq \sum_{a \neq a^*} \frac{8R^2 \log T}{\Delta_a}$$

On $\mathcal{B}$, since per-round regret is at most R: $\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{B}] \leq TR$
Final bound.

$$\mathbb{E}[\mathrm{Reg}(T)] \leq \sum_{a \neq a^*} \frac{8R^2 \log T}{\Delta_a} + |\mathcal{A}|R\,T^{-2}$$

Task 2: Implement UCB for Case B

Difference from previous case. Instead of treating each channel pair as an independent arm, the problem splits into two independent bandit problems. We therefore maintain two UCB indices, one for each user–channel pair, and at every round select the channels with the highest UCB values subject to the constraint $j_1 \neq j_2$, ensuring a non-conflicting pair.

Reward range. The user–channel component reward lies in the interval $R_i^t(j) \in [\log 3,\ \log 27]$, thus the reward range used in Hoeffding's inequality is: $R_i = \max R_i^t(j) - \min R_i^t(j) = \log 27 - \log 3 = \log 9$.

Choice of the UCB bonus term. The exploration bonus is chosen from Hoeffding's inequality. If the reward of the user–channel component lies in an interval of length R_i, then for any combination (i, j) and any number of pulls $N_{i,j}(t)$,

$$\Pr(|\hat{\mu}_{i,j}(t) - \mu_{i,j}| > \epsilon) \leq \exp\left(-\frac{2N_{i,j}(t)\epsilon^2}{R_i^2}\right)$$

We set the upper bound of the probability to T^{-4} and thus obtain: $\epsilon = R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}}$

UCB Policy. For $t = 1, \ldots, T$:

- For $t \leq K$, each user selects each channel exactly once.
- For $t > K$, compute $\mathrm{UCB}_{i,j}(t)$ for each user i and channel j: $\mathrm{UCB}_{i,j}(t) = \hat{\mu}_{i,j}(t) + R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}}$.
- Select the best non-conflicting pair: $(j_{1,t}, j_{2,t}) = \arg\max_{j_1 \neq j_2} (\mathrm{UCB}_{1,j_1}(t) + \mathrm{UCB}_{2,j_2}(t))$.
- Observe $R_1^t(j_{1,t}), R_2^t(j_{2,t})$ and update $N_{1,j_{1,t}}(t)$, $\hat{\mu}_{1,j_{1,t}}(t)$, $N_{2,j_{2,t}}(t)$, $\hat{\mu}_{2,j_{2,t}}(t)$.
- Track regret: $\mathbb{E}[\mathrm{Reg}(t)] = \mathbb{E}\left[\sum_{s=1}^{t}\left(\mu^* - \left(\mu_{1,j_{1,s}} + \mu_{2,j_{2,s}}\right)\right)\right]$, where $\mu^* = \max_{j_1 \neq j_2}\left(\mu_{1,j_1} + \mu_{2,j_2}\right)$.
  By defining the suboptimality gap $\Delta_{i,j} = \mu_{i,j_i^*} - \mu_{i,j}$ and rearranging the terms,

$$\mathbb{E}[\mathrm{Reg}(t)] = \mathbb{E}\left[\sum_{s=1}^{t}\left(\Delta_{1,j_{1,s}} + \Delta_{2,j_{2,s}}\right)\right] = \mathbb{E}\left[\sum_j \Delta_{1,j}N_{1,j}(t) + \sum_j \Delta_{2,j}N_{2,j}(t)\right] =$$
$$= \sum_{j \neq j_1^*} \Delta_{1,j}\,\mathbb{E}[N_{1,j}(t)] + \sum_{j \neq j_2^*} \Delta_{2,j}\,\mathbb{E}[N_{2,j}(t)]$$

Calculating instance-dependent bound

Step 1: Good event. Define the good event $\mathcal{G} := \left\{\forall i, j, t : |\hat{\mu}_{i,j}(t) - \mu_{i,j}| \leq R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}}\right\}$

Step 2: Pull bound on $\mathcal{G}$. Suppose channel $j \neq j_i^*$ is selected for user i at round t. Then by UCB,

$$\hat{\mu}_{i,j}(t) + R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}} \geq \hat{\mu}_{i,j_i^*}(t) + R_i\sqrt{\frac{2\log T}{N_{i,j_i^*}(t)}}$$

On $\mathcal{G}$,

$$\hat{\mu}_{i,j}(t) \leq \mu_{i,j} + R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}}, \quad \hat{\mu}_{i,j_i^*}(t) \geq \mu_{i,j_i^*} - R_i\sqrt{\frac{2\log T}{N_{i,j_i^*}(t)}}$$

Therefore,

$$\mu_{i,j} + 2R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}} \geq \mu_{i,j_i^*} \Leftrightarrow \Delta_{i,j} \leq 2R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}} \Leftrightarrow N_{i,j}(t) \leq \frac{8R_i^2\log T}{\Delta_{i,j}^2}$$

Step 3: Probability of $\mathcal{B}$. $\Pr(\mathcal{B}) = \Pr\left(\exists\, i, j, t : |\hat{\mu}_{i,j}(t) - \mu_{i,j}| > R_i\sqrt{\frac{2\log T}{N_{i,j}(t)}}\right) \leq 2KT^{-3}$

Step 4: Expected regret decomposition. $\mathbb{E}[\mathrm{Reg}(T)] = \Pr(\mathcal{G})\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{G}] + \Pr(\mathcal{B})\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{B}]$

On $\mathcal{G}$,

$$\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{G}] = \sum_{j\neq j_1^*}\Delta_{1,j}\mathbb{E}[N_{1,j}(T) \mid \mathcal{G}] + \sum_{j\neq j_2^*}\Delta_{2,j}\mathbb{E}[N_{2,j}(T) \mid \mathcal{G}] \leq \sum_{j\neq j_1^*}\frac{8R_1^2\log T}{\Delta_{1,j}} + \sum_{j\neq j_2^*}\frac{8R_2^2\log T}{\Delta_{2,j}}$$

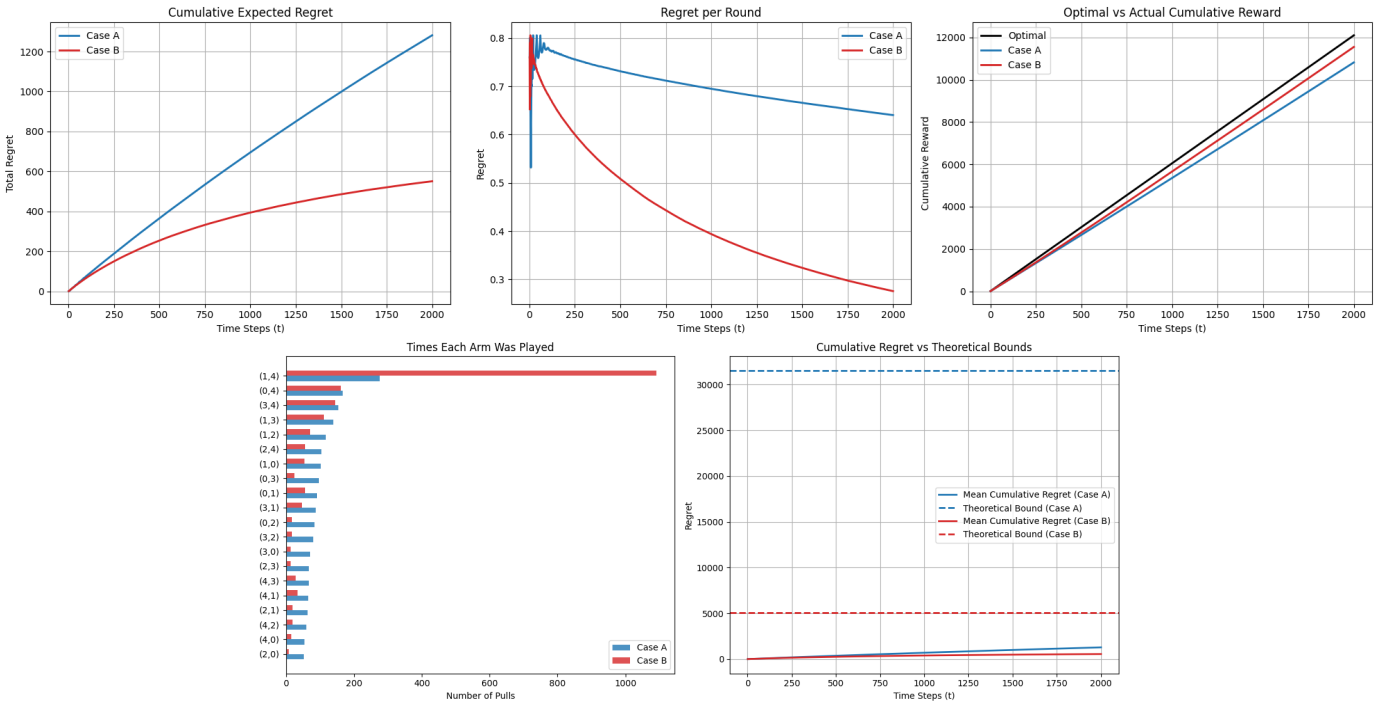
On $\mathcal{B}$, since the per-round regret is at most $R_1 + R_2$: $\mathbb{E}[\mathrm{Reg}(T) \mid \mathcal{B}] \leq T(R_1 + R_2)$

Final bound.

$$\mathbb{E}[\mathrm{Reg}(T)] \leq \sum_{j\neq j_1^*}\frac{8R_1^2\log T}{\Delta_{1,j}} + \sum_{j\neq j_2^*}\frac{8R_2^2\log T}{\Delta_{2,j}} + 2K(R_1 + R_2)\,T^{-2}$$

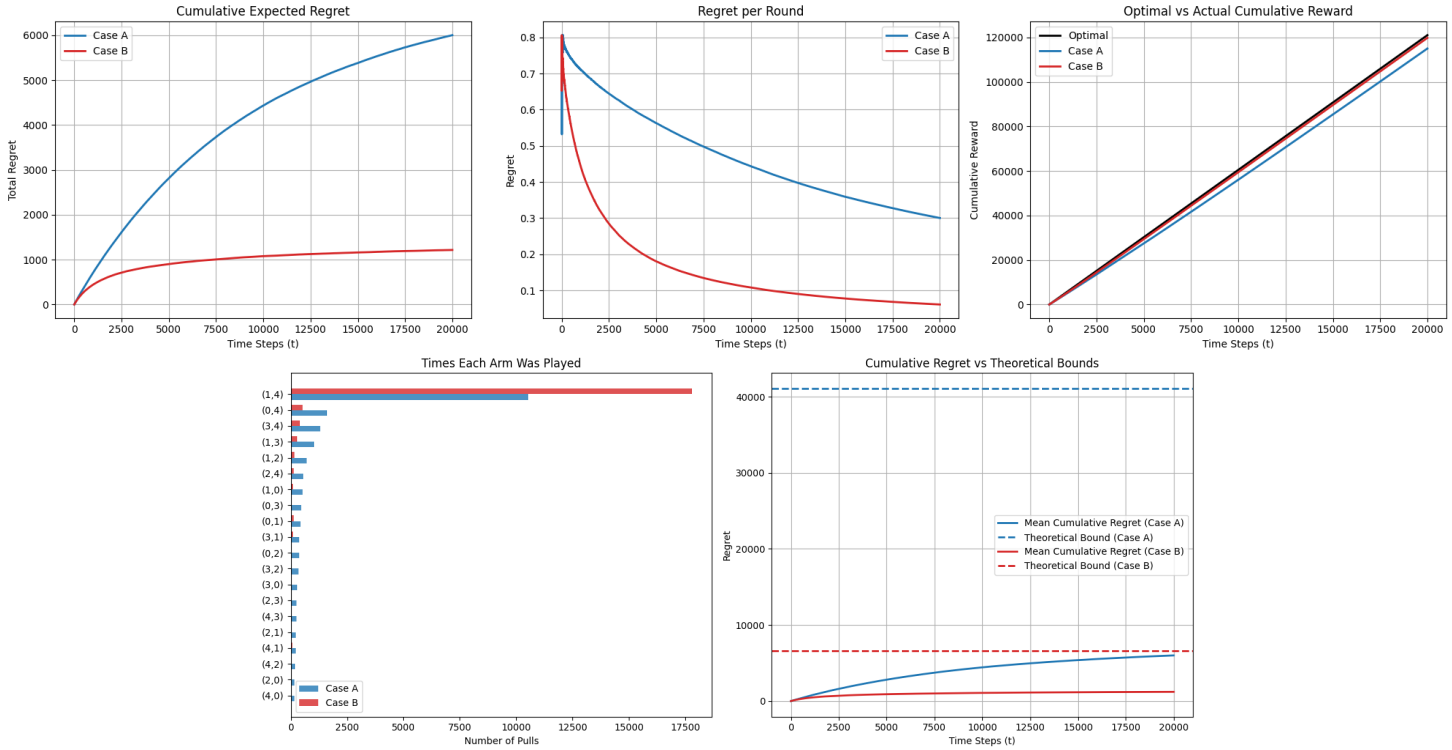
Task 3: Experimental comparison

▷ For the time horizon $T = 2000$, we obtain the following results:



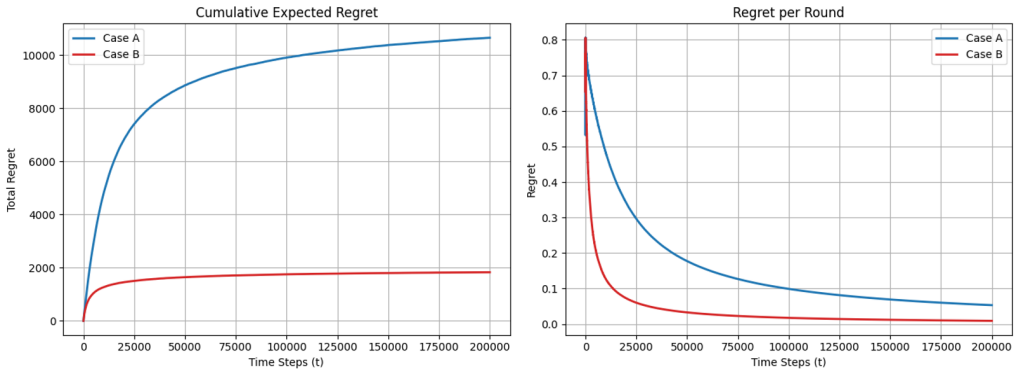
We can already observe that in both cases the algorithm is learning, as the regret per round decreases over time. At the same time, the cumulative regret is sublinear and remains significantly below the theoretical bound, confirming the validity of the derived regret guarantees. We can also already see what we expected: Case B learns faster, which is evident from the fact that it selects the optimal arm more frequently. This behavior is expected due to the additional structural information that the algorithm can exploit.

▷ For the time horizon $T = 20000$, we obtain the following results:



At this time horizon, the superiority of Case B becomes much clearer. The regret per round for Case B decreases much faster and tends toward zero, while Case A continues to converge at a significantly slower rate. The cumulative regret in both cases remains well below the theoretical bounds throughout the horizon.

▷ For the time horizon $T = 200000$, we obtain the following results:



Finally, by increasing the horizon even further, we observe that Case A requires roughly an order of magnitude larger horizon to reach regret-per-round levels comparable to those achieved by Case B much earlier. At this point, Case B has essentially zero regret per round and has fully learned the optimal arm.